

RAJALAKSHMI ENGINEERING COLLEGE

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CS23621

Mobile Application Development Laboratory



**RAJALAKSHMI
ENGINEERING
COLLEGE**

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)

CS23621 Mobile Application Development Laboratory LAB MANUAL

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Ex. No : 1

Kotlin Basics

Aim

To understand the basics of Kotlin programming including variables, data types, and simple input/output operations.

Procedure

1. Open Android Studio and create a new Kotlin project:
 - Launch Android Studio.
 - Click New Project → Empty Activity → Kotlin as the programming language.
 - Give your project a name (e.g., KotlinBasics).
 - This sets up the environment where you can write and run Kotlin programs.
2. Write a Kotlin program to demonstrate variables, constants, and data types:
 - In Kotlin, variables are used to store data.
 - Variables can be declared using var (mutable) or val (immutable/constant).
 - Kotlin is statically typed, which means the type of data is defined at compile time.
3. Run the program to check output:
 - Click the Run button in Android Studio.
 - Observe the output in the Run Console.
 - The program prints the values stored in variables to the console.

STEP 1: Open Android Studio

1. Open Android Studio (Otter / Otter 2)
2. Click New Project

STEP 2: Create New Project

1. Select Empty Activity
2. Click Next

Field	Value
Name	KotlinBasics
Language	Kotlin
Minimum SDK	API 21 or above
Build Configuration	View-based (NOT Compose)

STEP 3: Open Layout File (TextView UI)

app → res → layout → activity_main.xml

STEP 4: Add TextView

Create Layout XML (activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:gravity="center"

    android:padding="16dp">

    <TextView

        android:id="@+id/txtOutput"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Output will appear here"

        android:textSize="20sp" />

</LinearLayout>
```

STEP 5: Open Kotlin File

app → java → com.example.kotlinbasics → MainActivity.kt

MainActivity.kt

```
package com.example.kotlinbasics

import android.os.Bundle

import android.widget.TextView

import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)

        // Reference to TextView

        val output = findViewById<TextView>(R.id.txtOutput)

        // Kotlin Basics

        val name = "Android"

        val a = 10

        val b = 20

        val sum = a + b

        // Display output

        output.text = "Welcome $name\nSum = $sum"

    }

}
```

STEP 7: Create Emulator

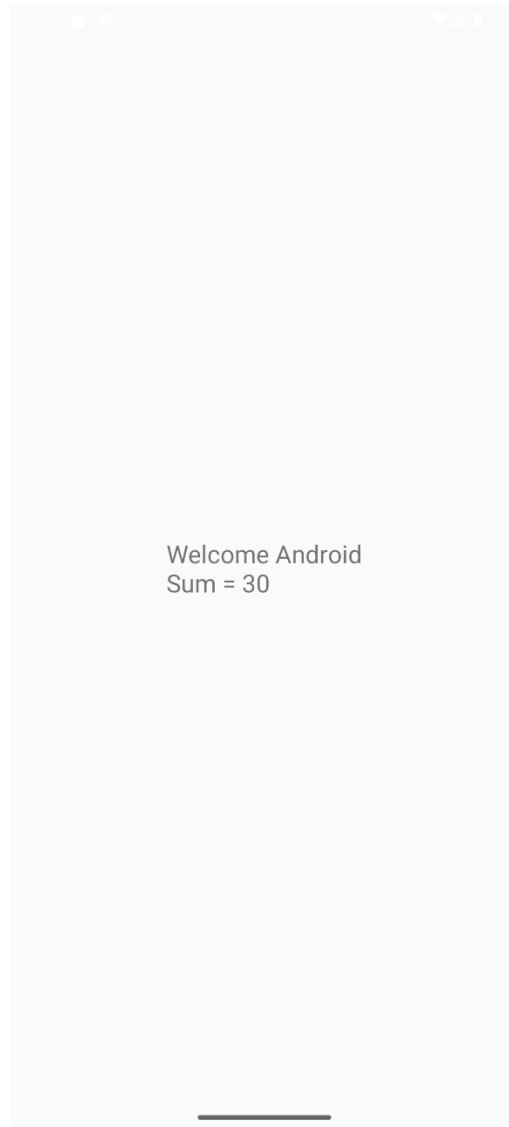
- Click Device Manager
- Click Create Device
- Choose Pixel / Medium Phone
- Select Android version
- Click Finish
- Start emulator 

STEP 8: Run the App

1. Select emulator
2. Click Run 

App opens on emulator

Output



Functions

Aim

To understand and implement **functions in Kotlin**, including **defining functions with parameters, return types, and calling them**, and display output in an Android app.

Explanation

- A function is a block of code designed to perform a specific task.
- In Kotlin, functions are defined using the fun keyword.
- Functions can:
 1. Take parameters (inputs).
 2. Return a value (Int, String, etc.) or Unit (no return).
 3. Be called multiple times to reuse code.
- In Android apps, all code must be inside a class (usually MainActivity) and runtime code goes inside onCreate().
- Output is displayed using a TextView, instead of println() like in console programs.

Procedure

1. Open Android Studio .
2. Open the project and go to res/layout/activity_main.xml.
3. Add a TextView to display output.
4. Open MainActivity.kt and:
 - Define functions inside the class.
 - Call the functions inside onCreate().
 - Set the output to the TextView.
5. Run the app on an emulator or physical device.
6. Observe the output displayed on the screen.

Notes :

1. All functions are inside the MainActivity class.
2. TextView is used to display output instead of println().
3. Functions demonstrate:
 - Parameter passing (add(a, b), greet(name))
 - Returning values (Int, String)
 - Function without parameters (welcomeMessage())

Code

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical"

    android:gravity="center"

    android:padding="16dp">

    <TextView

        android:id="@+id/txtOutput"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Kotlin Output"

        android:textSize="20sp" />

</LinearLayout>
```

MainActivity.kt

```
package com.example.kotlinbasics

import android.os.Bundle

import android.widget.TextView

import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)

        val txt = findViewById<TextView>(R.id.txtOutput)

        val a = 10

        val b = 20

        val sum = a + b

        txt.text = "Welcome to Kotlin\nSum = $sum"

    }

}
```

Output



Classes and Objects

Aim

To understand the concept of **classes and objects in Kotlin**, including **creating classes, defining properties and methods**, and using **objects to access class members** in an Android app.

Explanation

- A class is a blueprint for creating objects.
- An object is an instance of a class.
- Classes can have:
 1. Properties (variables inside class)
 2. Methods/Functions (actions the class can perform)

In Kotlin:

```
class Person(val name: String, var age: Int) {  
  
    fun greet() = "Hello, $name!"  
  
}
```

In Android, we create objects inside MainActivity and use a TextView to display output.

Procedure :

1. Open activity_main.xml and ensure there is a TextView for output:
2. Open MainActivity.kt.
3. Define a class inside the Kotlin file or in a separate file.
4. Create an object of the class inside onCreate().
5. Call class methods and display output in the TextView.
6. Run the app to see the results

Notes

1. Person class has:
 - Property name (immutable)
 - Property age (mutable)
 - Method greet() returns a greeting
 - Method birthday() updates age
2. Object person is used to access class properties and methods.
3. Output is displayed in TextView, suitable for Android apps.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:orientation="vertical">

    <!-- App Toolbar -->

    <com.google.android.material.appbar.MaterialToolbar

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:title="Kotlin Classes"

        android:titleTextColor="@android:color/white"

        android:background="?attr/colorPrimary" />

    <!-- Main Content -->

    <LinearLayout

        android:layout_width="match_parent"

        android:layout_height="match_parent"

        android:orientation="vertical"

        android:gravity="center"
```

android:padding="24dp"

android:background="#FAFAFA">

<Button

android:id="@+id/btnShow"

android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:text="Show Student Details" />

<TextView

android:id="@+id/txtOutput"

android:layout_width="wrap_content"

android:layout_height="wrap_content"

android:text="Output will appear here"

android:textSize="18sp"

android:padding="16dp"

android:layout_marginTop="16dp"

android:background="@android:color/white"

android:elevation="6dp" />

</LinearLayout>

</LinearLayout>

Create a Kotlin Class

- Right-click package name
- Select New → Kotlin Class/File
- Name it: Student
- Choose Class

Write Student.kt

```
package com.example.kotlinbasics

class Student(

    val name: String,

    val rollNo: Int

) {

    fun displayDetails(): String {

        return "Name: $name\nRoll No: $rollNo"

    }

}
```

Update strings.xml

res → values → strings.xml

```
<string name="student_output">Student Details\n%1$s</string>
```

Code

MainActivity.kt

```
package com.example.kotlinbasics

import android.os.Bundle

import android.widget.Button

import android.widget.TextView

import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {
```



```
override fun onCreate(savedInstanceState: Bundle?) {  
  
    super.onCreate(savedInstanceState)  
  
    setContentView(R.layout.activity_main)  
  
  
    val btn = findViewById<Button>(R.id.btnShow)  
  
    val txt = findViewById<TextView>(R.id.txtOutput)  
  
  
    btn.setOnClickListener {  
  
        // Creating object of Student class  
  
        val student = Student("Arun", 101)  
  
  
        txt.text = getString(  
  
            R.string.student_output,  
  
            student.displayDetails()  
  
        )  
  
    }  
  
}  
  
}
```

STEP 7: Run the App

1. Start emulator
2. Click Run 
3. Tap Show Student Details

Output :



Build Your First Android App

Aim

To create a simple Android app using Kotlin, understand Activity, layout, and display output on screen.

Prerequisites:

1. Android Studio installed.
2. Kotlin selected as the programming language.
3. Basic understanding of Activities and Layouts.

Explanation

- An Android app is composed of:
 1. Activities – Screens that users interact with (MainActivity)
 2. Layout XML – Defines the UI components (TextView, Button, etc.)
 3. Manifest – Declares the app components and permissions (AndroidManifest.xml)
- MainActivity is the entry point of the app.
- setContentView(R.layout.activity_main) connects the Activity with its layout.
- Output is displayed on TextView in the UI instead of console output.

Procedure

1. Open Android Studio Otter 2.
2. Create a new project → Empty Activity → Kotlin → Finish.
3. Open res/layout/activity_main.xml and add a TextView:
4. Open MainActivity.kt.
5. Inside onCreate(), find the TextView using findViewById and set its text.
6. Run the app on an **emulator or physical device**.
7. Observe the output on the app screen.

Step 1: Create a New Project

1. Open **Android Studio** → Click **“New Project”**.
2. Select **Empty Activity** → Click **Next**.
3. Give your project a name, e.g., MyFirstApp.
4. Set **Language** = **Kotlin**.
5. Minimum SDK → Choose **API 21: Android 5.0 (Lollipop)** or higher.
6. Click **Finish**.

Step 2: Design the Layout

1. Open **res/layout/activity_main.xml**.

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:orientation="vertical"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:padding="16dp"
```

```
    android:gravity="center">
```

```
    <TextView
```

```
        android:id="@+id/tvMessage"
```

```
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
```

```
        android:text="Hello, Android!"
```

```
        android:textSize="24sp"
```

```
        android:textColor="#000000"
```

```
        android:padding="16dp"/>
```

```
<Button

    android:id="@+id/btnClick"

    android:layout_width="wrap_content"

    android:layout_height="wrap_content"

    android:text="Click Me"/>

</LinearLayout>
```

Step 3: Add Logic in Kotlin

Code

MainActivity.kt

```
package com.example.myfirstapp

import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

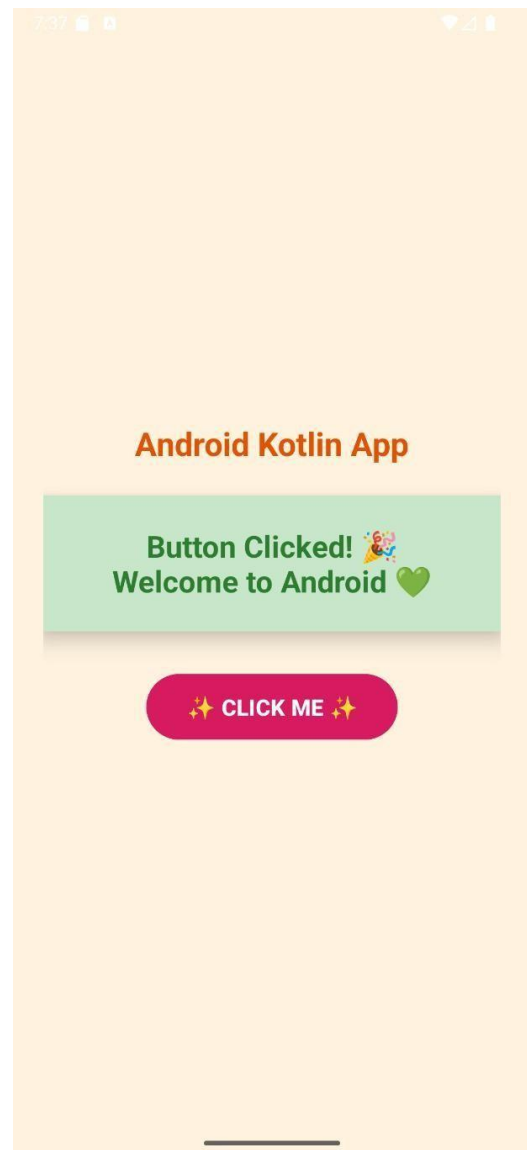
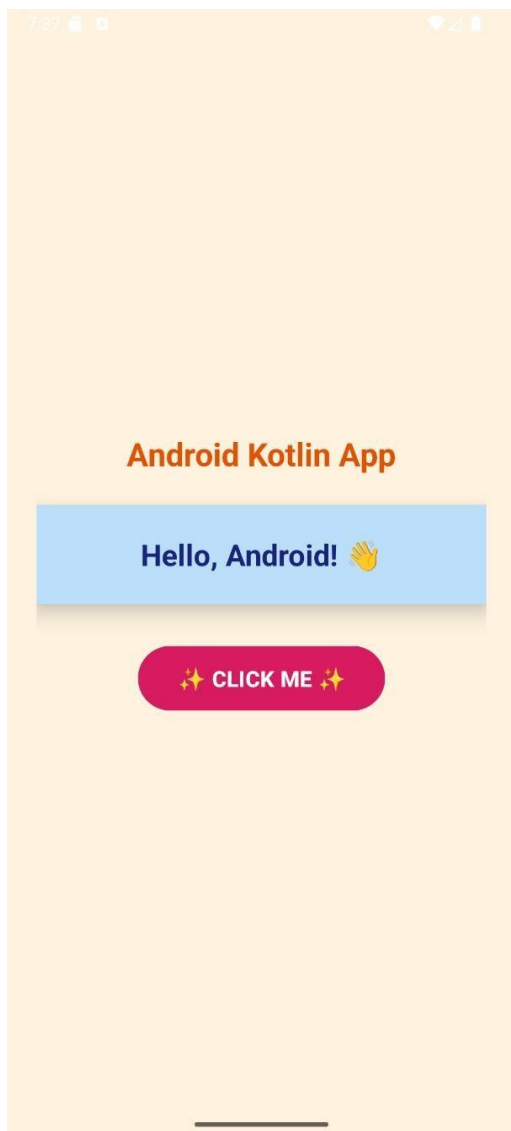
        val tvMessage = findViewById<TextView>(R.id.tvMessage)
        val btnClick = findViewById<Button>(R.id.btnClick)

        btnClick.setOnClickListener {
            tvMessage.text = "Button Clicked! Welcome to Android."
        }
    }
}
```

Step 4: Run the App

1. Connect an Android device or use an emulator.
2. Click Run () in Android Studio.
3. You should see:
 - Initial text: Hello, Android!
 - On button click → Text changes to Button Clicked! Welcome to Android.

Output



Aim

To understand different types of layouts in Android, including LinearLayout, RelativeLayout, and ConstraintLayout, and design user interfaces.

Android Layouts define how UI elements are arranged on the screen.

Common layouts:

1. LinearLayout – Arranges elements vertically or horizontally.
2. RelativeLayout – Elements positioned relative to each other or parent.
3. ConstraintLayout – Modern, flexible layout with constraints.
4. FrameLayout – Stack elements on top of each other.
5. TableLayout – Arrange elements in rows and columns.

Procedure:

1. Open activity_main.xml.
2. Design UI using LinearLayout with multiple TextViews, Buttons, or ImageViews.
3. Experiment with orientation (vertical/horizontal) and padding/margin.
4. Try RelativeLayout to position elements relative to each other.
5. Try ConstraintLayout to create more flexible layouts using constraints.
6. Run the app on an emulator/device and observe the UI.

Step 1: Create a New Project

- Open Android Studio → New Project → Empty Activity
- Name: LayoutsEx5
- Language: Kotlin
- Minimum API: 21+
- Click Finish

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:padding="16dp">
    <TextView
        android:id="@+id/tvWelcome"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Layouts Ex-5"
        android:textSize="24sp"
        android:textStyle="bold" />
    <Button
        android:id="@+id/btnNext"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Go to Second Activity"
        android:layout_marginTop="20dp" />

</LinearLayout>
```


Step 3: Create Second Activity

- Right-click app → New → Activity → Empty Activity
- Name: SecondActivity
- Ensure Generate Layout File is checked → Click Finish

Step 4: Design SecondActivity Layout

File: res/layout/activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:orientation="vertical"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:gravity="center"
```

```
    android:padding="16dp">
```

```
    <TextView
```

```
        android:id="@+id/tvSecond"
```

```
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
```

```
        android:text="This is the Second Activity"
```

```
        android:textSize="24sp"
```

```
        android:textStyle="bold" />
```

```
</LinearLayout>
```

Step 5: Add Button Functionality in MainActivity

MainActivity.kt

```
package com.example.layoutsex5

import android.content.Intent
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        // Find the button by ID
        val btnNext = findViewById<Button>(R.id.btnNext)

        // Set OnClickListener
        btnNext.setOnClickListener {
            // Navigate to SecondActivity
            val intent = Intent(this, SecondActivity::class.java)
            startActivity(intent)
        }
    }
}
```

Step 6: Ensure SecondActivity Code

File: SecondActivity.kt

```
package com.example.layoutsex5

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle

class SecondActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_second)

    }

}
```

Step 7: Update AndroidManifest

```
<application

    android:allowBackup="true"

    android:label="@string/app_name"

    android:theme="@style/Theme.LayoutsEx5">

    <activity android:name=".SecondActivity"/>

    <activity android:name=".MainActivity">

        <intent-filter>

            <action android:name="android.intent.action.MAIN"/>

            <category android:name="android.intent.category.LAUNCHER"/>

        </intent-filter>

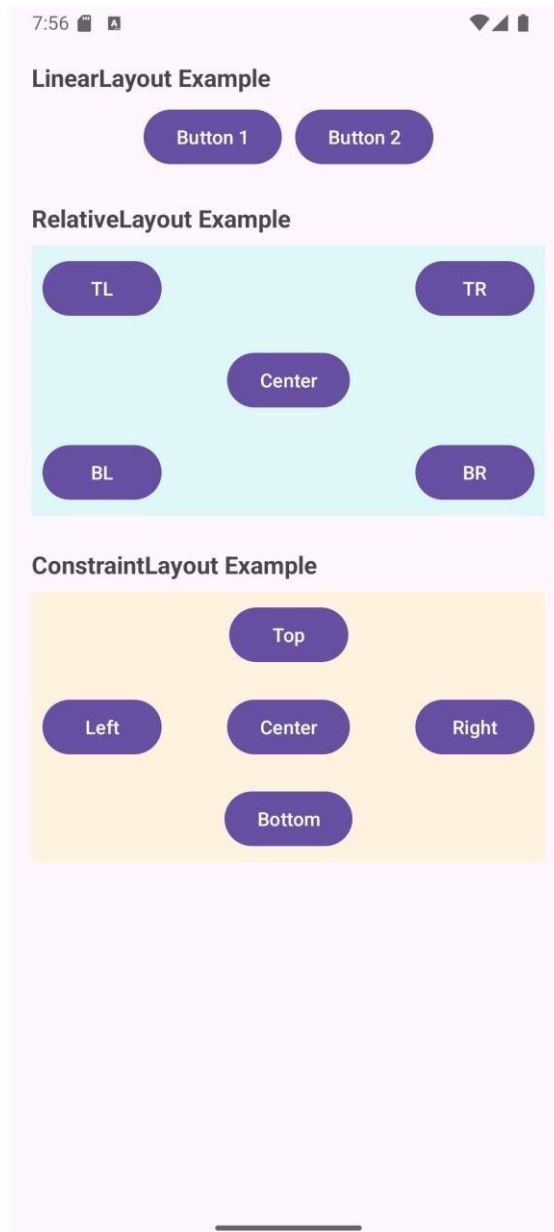
    </activity>

</application>
```

Step 9: Run the App

- Connect a device/emulator
- Click Run → Run 'app'
- Check the output:
 1. MainActivity shows TextView + Button
 2. Click button → opens SecondActivity

Output



App Navigation

Aim

To understand navigation between multiple screens in an Android application using **Intents** and **multiple Activities**.

Explanation :

- In Android, an Activity represents a single screen with a user interface.
- Most Android applications consist of multiple screens, such as:
 - Login screen
 - Home screen
 - Details screen

Why Navigation is Needed

- Navigation allows users to move from one screen to another.
- Android uses an Intent to perform this task.

Intent – Core Concept

- An Intent is a messaging object used to:
 - Start another Activity
 - Pass data between Activities
- There are two types of intents:
 - Explicit Intent – Used to move between activities within the same app
 - Implicit Intent – Used to perform system actions (call, open browser, etc.)

How Navigation Works

1. User clicks a Button in the first Activity.
2. OnClickListener listens for the click event.
3. An Intent object is created with:
 - Current Activity
 - Target Activity
4. startActivity() launches the new screen.

Procedure

1. Open Android Studio and create a project using Empty Activity.
2. Design the first screen with a Button.
3. Create a Second Activity.
4. Add a TextView to the second Activity layout.
5. Write Kotlin code to handle Button click.
6. Use an Intent to navigate from the first Activity to the second Activity.
7. Run the app and test navigation.

Step 1: Create a New Project

1. Open Android Studio → New Project → Empty Activity.
2. Name the project: AppNavigationDemo.
3. Language: Kotlin. Minimum SDK: API 21 or higher.
4. Click Finish.

Step 2: Create a Second Activity

1. Right-click on app → java → <your package> → New → Activity → Empty Activity.
2. Name it SecondActivity.
3. Click Finish.

Step 3: Design Layouts

1. MainActivity Layout (activity_main.xml)

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:orientation="vertical"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:gravity="center"
```

```
    android:padding="16dp">
```

```
    <TextView
```

```
        android:id="@+id/tvMain"
```

```
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
```

```
        android:text="Welcome to Main Activity"
```

```
        android:textSize="24sp"
```

```
        android:padding="16dp"/>
```

```
    <Button
```

```
        android:id="@+id/btnNext"
```

```
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
```

```
        android:text="Go to Second Activity"/>
```

```
</LinearLayout>
```

2. SecondActivity Layout (activity_second.xml)

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:padding="16dp">

    <TextView
        android:id="@+id/tvSecond"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Second Activity"
        android:textSize="24sp"
        android:padding="16dp"/>

</LinearLayout>
```

Step 4: Add Navigation Logic in Kotlin

1. MainActivity.kt

```
package com.example.appnavigationdemo

import android.content.Intent
import android.os.Bundle
import android.widget.Button
import androidx.appcompat.app.AppCompatActivity
```



```
class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val btnNext = findViewById<Button>(R.id.btnNext)

        btnNext.setOnClickListener {
            // Intent to navigate to SecondActivity
            val intent = Intent(this, SecondActivity::class.java)
            startActivity(intent)
        }
    }
}
```

2. SecondActivity.kt

```
package com.example.appnavigationdemo

import android.os.Bundle
import androidx.appcompat.app.AppCompatActivity

class SecondActivity : AppCompatActivity() {

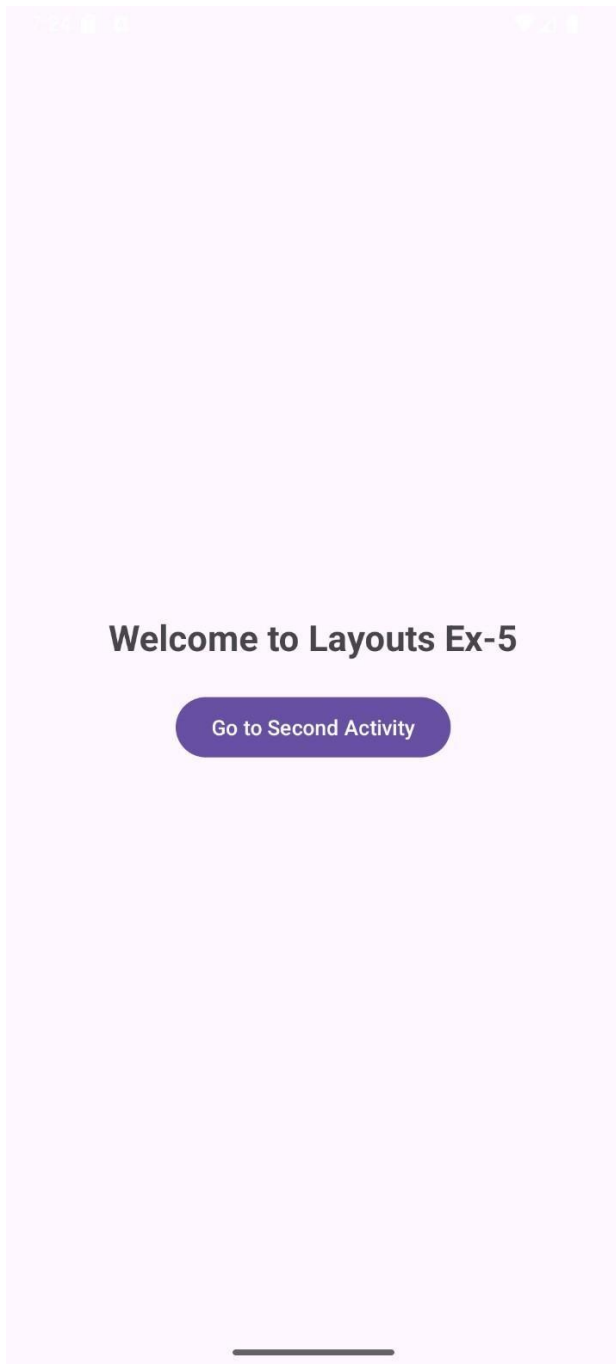
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_second)
    }
}
```

Step 5: Run the App

1. Connect an Android device or use an emulator.
2. Click Run () in Android Studio.
3. You should see MainActivity.
4. Click “Go to Second Activity” → SecondActivity appears.

Output

- When the application starts, Main Activity is displayed.
- Clicking the “Go to Second Screen” button opens Second Activity.
- The second screen displays the message:



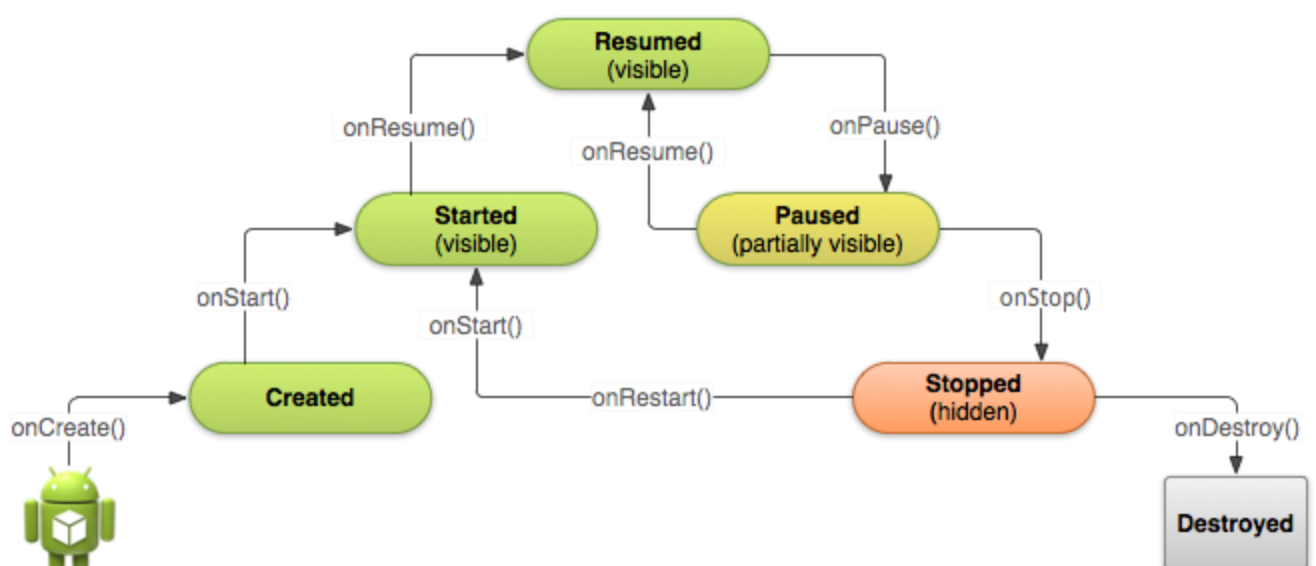
Activity and Fragment Lifecycles

Aim

To understand the **lifecycle of an Activity and Fragment** in Android and observe how different lifecycle methods are invoked during various states of the application.

What is Lifecycle?

- The lifecycle refers to the different states an Activity or Fragment goes through from creation to destruction.
- Android automatically calls lifecycle methods based on user actions, such as:
 - Opening the app
 - Switching apps
 - Rotating the device
 - Closing the app



Why Lifecycle Management is Important

- To manage resources efficiently
- To save and restore user data
- To avoid memory leaks
- To ensure smooth app behavior during background/foreground transitions

Activity Lifecycle Methods

Method	Purpose
onCreate()	Activity is created
onStart()	Activity becomes visible
onResume()	Activity starts interacting with user
onPause()	Activity partially hidden
onStop()	Activity no longer visible
onDestroy()	Activity is destroyed

Fragment Lifecycle (Brief)

- Fragment lifecycle is similar to Activity but depends on the host Activity.
- Important methods include:
 - onCreateView()
 - onStart()
 - onResume()
 - onPause()
 - onStop()
 - onDestroyView()

How Lifecycle is Observed

- Lifecycle events are observed using:
 - Logcat messages
 - TextView updates

Procedure

1. Create an Android project using Empty Activity.
2. Open MainActivity.kt.
3. Override Activity lifecycle methods.
4. Add Log statements in each lifecycle method.
5. Run the app.
6. Perform actions:
 - Open app
 - Press Home button
 - Reopen app
 - Rotate screen
 - Close app
7. Observe lifecycle callbacks in Logcat.

STEP 1: Create New Project

1. Open Android Studio
2. Click New Project
3. Select Empty Views Activity
4. Click Next
5. Enter:
 - Name: LifecycleDemo
 - Language: Kotlin
 - Minimum SDK: API 21
6. Click Finish
7. Wait until Build Successful

STEP 2: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"

    android:orientation="vertical"

    android:layout_width="match_parent"

    android:layout_height="match_parent">

    <FrameLayout

        android:id="@+id/fragment_container"

        android:layout_width="match_parent"

        android:layout_height="0dp"

        android:layout_weight="1"/>

    <TextView

        android:id="@+id/tvLifecycle"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Lifecycle Output:"

        android:textSize="16sp"

        android:padding="10dp"/>

</LinearLayout>
```

STEP 3: Edit MainActivity.kt

```
package com.example.lifecycledemo

import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    lateinit var tvLifecycle: TextView

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        tvLifecycle = findViewById(R.id.tvLifecycle)
        tvLifecycle.append("\nActivity onCreate")

        supportFragmentManager.beginTransaction()
            .replace(R.id.fragment_container, DemoFragment())
            .commit()
    }

    override fun onStart() {
        super.onStart()
        tvLifecycle.append("\nActivity onStart")
    }
}
```



```
override fun onResume() {  
    super.onResume()  
    tvLifecycle.append("\nActivity onResume")  
}
```

```
override fun onPause() {  
    super.onPause()  
    tvLifecycle.append("\nActivity onPause")  
}
```

```
override fun onStop() {  
    super.onStop()  
    tvLifecycle.append("\nActivity onStop")  
}
```

```
override fun onDestroy() {  
    super.onDestroy()  
    tvLifecycle.append("\nActivity onDestroy")  
}  
}
```

STEP 4: Create Fragment Class

1. Right-click package name
2. Select New → Kotlin Class/File
3. Name: DemoFragment
4. Type: Class
5. Click OK

STEP 5: Write DemoFragment.kt Code

```
package com.example.lifecycledemo

import android.os.Bundle
import android.view.LayoutInflater
import android.view.View
import android.view.ViewGroup
import androidx.fragment.app.Fragment

class DemoFragment : Fragment() {

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)

        (activity as MainActivity).tvLifecycle.append("\nFragment onCreate")
    }

    override fun onCreateView(

        inflater: LayoutInflater,

        container: ViewGroup?,

        savedInstanceState: Bundle?

    ): View? {

        (activity as MainActivity).tvLifecycle.append("\nFragment onCreateView")

        return inflater.inflate(R.layout.fragment_demo, container, false)
    }

    override fun onResume() {

        super.onResume()

        (activity as MainActivity).tvLifecycle.append("\nFragment onResume")
    }
}
```

```

override fun onPause() {

    super.onPause()

    (activity as MainActivity).tvLifecycle.append("\nFragment onPause")

}

override fun onDestroyView() {

    super.onDestroyView()

    (activity as MainActivity).tvLifecycle.append("\nFragment onDestroyView")

}

override fun onDetach() {

    super.onDetach()

    (activity as MainActivity).tvLifecycle.append("\nFragment onDetach")

}

}

```

STEP 6: Create Fragment Layout

res → layout → New → Layout Resource File

- File name: fragment_demo
- Root element: TextView

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<TextView
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:text="Fragment Lifecycle Demo"
```

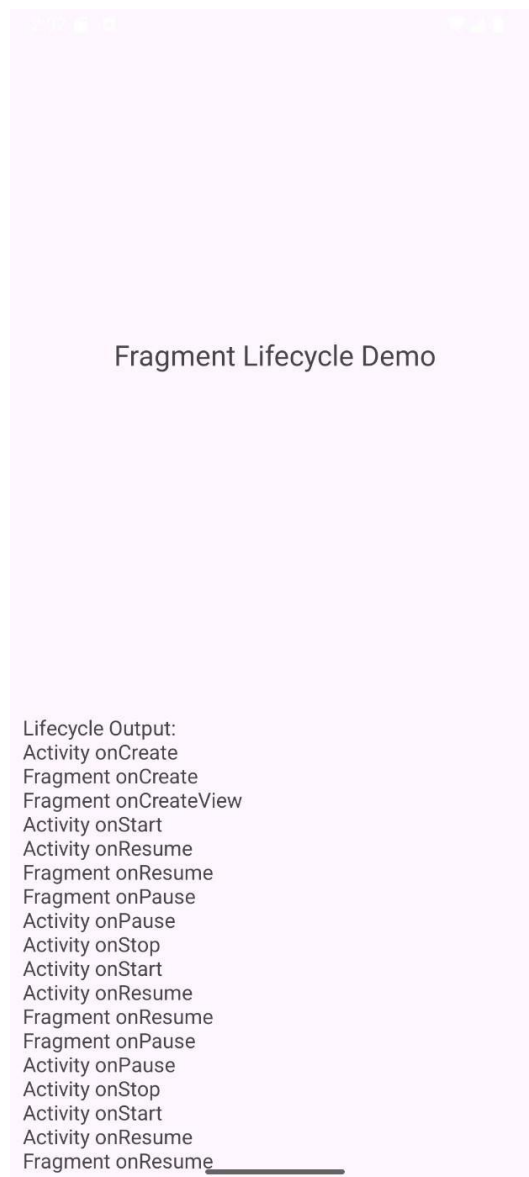
```
    android:textSize="22sp"
```

```
    android:gravity="center"/>
```

STEP 7: Run Using Device Manager

- Click Device Manager
- Start an Emulator
- Click Run ►
- App opens on device

Output



Ex. No : 8

App Architecture – UI Layer

Aim

To understand the **UI layer architecture** in Android applications and implement user interaction by separating UI components and event-handling logic.

What is App Architecture?

- App architecture defines how different parts of an application are organized and interact with each other.
- A well-structured architecture makes the app:
 - Easy to understand
 - Easy to maintain
 - Easy to extend

UI Layer – Core Concept

- The UI Layer is responsible for:
 - Displaying data to the user
 - Handling user interactions (clicks, inputs)
- It consists of:
 - XML Layouts → UI design
 - Activity / Fragment → UI logic

Responsibilities of UI Layer

- Capture user actions (Button click, text input)
- Update UI elements dynamically
- Communicate with underlying logic (future ViewModel / data layer)

Procedure

1. Create a new Android project using Empty Activity.
2. Design the UI in activity_main.xml using TextView, EditText, and Button.
3. Open MainActivity.kt.
4. Handle user input and Button click inside the Activity.
5. Update TextView dynamically based on user action.
6. Run the app and observe UI behavior.

STEP 1: Create New Project

1. Open Android Studio
2. Click New Project
3. Select Empty Views Activity
4. Click Next
5. Enter:
 - Name: UILayerDemo
 - Language: Kotlin
 - Min SDK: API 21
6. Click Finish

STEP 2: Design UI (XML)

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:orientation="vertical"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:padding="20dp">
```

<TextView

android:id="@+id/tvTitle"

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:text="UI Layer Demo"

android:textSize="22sp"

android:gravity="center"/>

<TextView

android:id="@+id/tvMessage"

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:text="Message will appear here"

android:textSize="18sp"

android:paddingTop="20dp"/>

<Button

android:id="@+id/btnLoad"

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:text="Load Data"/>

</LinearLayout>

STEP 3: Create ViewModel

1. Right-click package name
2. Select New → Kotlin Class/File
3. Name: MainViewModel
4. Type: Class

```
package com.example.uilayerdemo
```

```
import androidx.lifecycle.ViewModel
```

```
class MainViewModel : ViewModel() {  
  
    fun getMessage(): String {  
  
        return "Hello from ViewModel (UI Layer)"  
  
    }  
  
}
```

STEP 4: MainActivity.kt

```
package com.example.uilayerdemo
```

```
import android.os.Bundle
```

```
import android.widget.Button
```

```
import android.widget.TextView
```

```
import androidx.activity.viewModels
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {  
  
    private val viewModel: MainViewModel by viewModels()  
  
    override fun onCreate(savedInstanceState: Bundle?) {  
  
        super.onCreate(savedInstanceState)  
  
        setContentView(R.layout.activity_main)
```



```
val tvMessage = findViewById<TextView>(R.id.tvMessage)
```

```
val btnLoad = findViewById<Button>(R.id.btnLoad)
```

```
btnLoad.setOnClickListener {
```

```
    tvMessage.text = viewModel.getMessage()
```

```
}
```

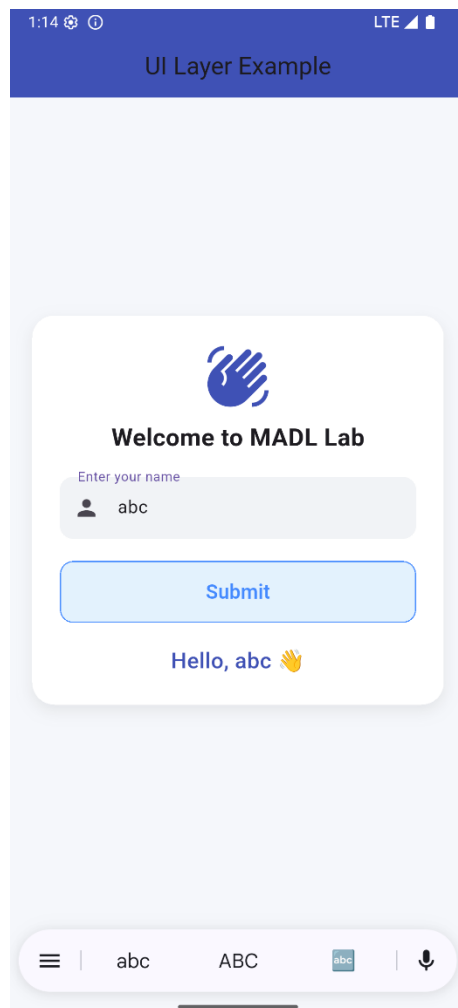
```
}
```

```
}
```

STEP 5: Run Using Device Manager

1. Open Device Manager
2. Start Emulator
3. Click Run

Output



Ex. No :09

Aim

To understand the Persistence Layer in Android application architecture and implement data storage using SharedPreferences.

Explanation

What is App Architecture?

- App architecture organizes an application into layers so that each layer has a specific responsibility.
- Common layers include:
 - UI Layer
 - Business Logic Layer
 - Persistence (Data) Layer

Persistence Layer – Core Concept

- The Persistence Layer is responsible for:
 - Storing data permanently
 - Retrieving stored data when needed
- Data remains available even after:
 - App is closed
 - Device is restarted

Why Persistence is Important

- To save user information (name, settings)
- To maintain application state
- To improve user experience by avoiding repeated data entry

SharedPreferences

- SharedPreferences is a lightweight storage mechanism in Android.
- It stores data as key–value pairs.
- Suitable for storing small amounts of simple data such as:
 - Username
 - Login status
 - App preferences

Working of SharedPreferences

1. User enters data.
2. Data is stored using a unique key.
3. Data can be retrieved using the same key.
4. Stored data persists across app restarts.

Procedure

1. Create a new Android project using Empty Activity.
2. Design UI with EditText, Buttons, and TextView.
3. Use SharedPreferences to save user data.
4. Retrieve and display saved data.
5. Close and reopen the app to verify persistence.

Persistence Layer Components

1. Entity – Represents a table
2. DAO – Defines database operations
3. Database – Holds the database

Why Room Database?

- Official Android library
- Easy to use
- Avoids raw SQL errors
- Lifecycle aware

STEP 1: Create New Project

1. Open Android Studio
2. Click New Project
3. Choose Empty Views Activity
4. Click Next
5. Enter:
 - Name: PersistenceDemo
 - Language: Kotlin
 - Min SDK: API 21
6. Click Finish

STEP 2: Add Room Dependencies

Gradle Scripts → build.gradle (Module :app)

build.gradle (Module :app)

```
plugins {  
    alias(libs.plugins.android.application)  
    alias(libs.plugins.kotlin.android)  
    alias(libs.plugins.kotlin.kapt)  
}
```

STEP 4: Dependencies block (keep as-is)

```
dependencies {  
    implementation(libs.androidx.room.runtime)  
    implementation(libs.androidx.room.ktx)  
    kapt(libs.androidx.room.compiler)  
}
```

STEP 5: Sync + Clean (IMPORTANT ORDER)

1. File → Sync Project with Gradle Files

activity main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"

    android:orientation="vertical"

    android:padding="20dp"

    android:layout_width="match_parent"

    android:layout_height="match_parent">

    <EditText

        android:id="@+id/etName"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:hint="Enter Name"/>
```

<Button

android:id="@+id/btnSave"

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:text="Save to Database"/>

<TextView

android:id="@+id/tvResult"

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:text="Stored Data:"

android:paddingTop="20dp"

android:textSize="18sp"/>

</LinearLayout>

STEP 4: Create Entity Class

Right-click package → New → Kotlin Class/File

Name: UserEntity

```
package com.example.persistencedemo
```

```
import androidx.room.Entity
```

```
import androidx.room.PrimaryKey
```

```
@Entity(tableName = "user_table")
```

```
data class UserEntity(
```

```
    @PrimaryKey(autoGenerate = true)
```

```
    val id: Int = 0,
```

```
    val name: String
```

```
)
```

STEP 5: Create DAO Interface

New → Kotlin Class/File

Name: UserDao

Type: Interface

```
package com.example.persistencedemo
```

```
import androidx.room.Dao
```

```
import androidx.room.Insert
```

```
import androidx.room.Query
```

```
@Dao
```

```
interface UserDao {
```

```
    @Insert
```

```
    suspend fun insertUser(user: UserEntity)
```

```
    @Query("SELECT * FROM user_table")
```

```
    suspend fun getAllUsers(): List<UserEntity>
```

```
}
```

STEP 6: Create Database Class

```
package com.example.persistencedemo

import android.content.Context
import androidx.room.Database
import androidx.room.Room
import androidx.room.RoomDatabase

@Database(entities = [UserEntity::class], version = 1)
abstract class AppDatabase : RoomDatabase() {

    abstract fun userDao(): UserDao

    companion object {

        private var INSTANCE: AppDatabase? = null

        fun getDatabase(context: Context): AppDatabase {

            return INSTANCE ?: synchronized(this) {

                Room.databaseBuilder(

                    context.applicationContext,

                    AppDatabase::class.java,

                    "user_database"

                ).build().also {

                    INSTANCE = it

                }

            }

        }

    }

}
```



```
    }  
}  
}
```

MainActivity.kt

```
package com.example.persistencedemo
```

```
import android.os.Bundle
```

```
import android.widget.Button
```

```
import android.widget.EditText
```

```
import android.widget.TextView
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
import kotlinx.coroutines.CoroutineScope
```

```
import kotlinx.coroutines.Dispatchers
```

```
import kotlinx.coroutines.launch
```

```
class MainActivity : AppCompatActivity() {
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
```

```
        setContentView(R.layout.activity_main)
```

```
        val etName = findViewById<EditText>(R.id.etName)
```

```
        val btnSave = findViewById<Button>(R.id.btnSave)
```

```
        val tvResult = findViewById<TextView>(R.id.tvResult)
```

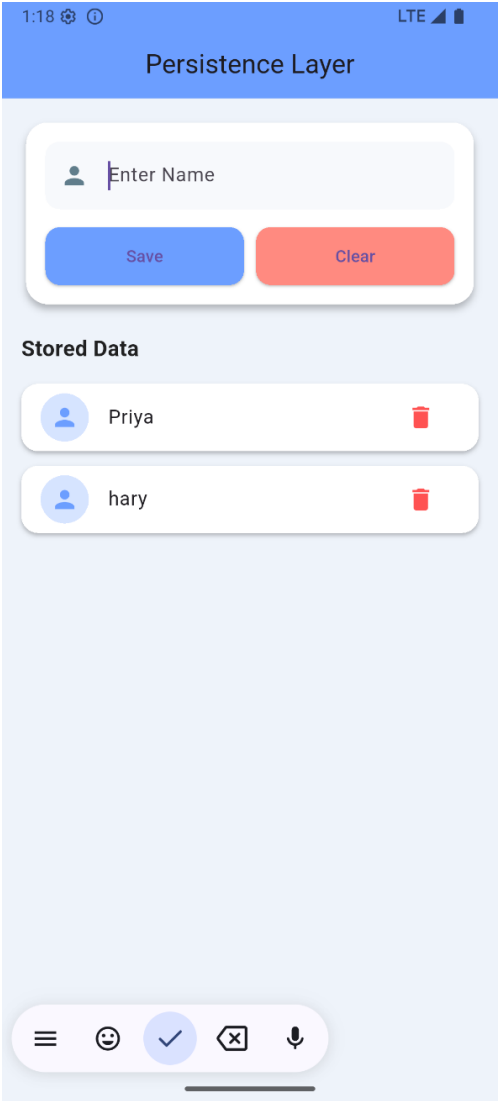
```
        val db = AppDatabase.getDatabase(this)
```

```
btnSave.setOnClickListener {  
    val name = etName.text.toString()  
  
    CoroutineScope(Dispatchers.IO).launch {  
        db.userDao().insertUser(UserEntity(name = name))  
        val users = db.userDao().getAllUsers()  
  
        runOnUiThread {  
            tvResult.text = "Stored Data:\n"  
            users.forEach {  
                tvResult.append(it.name + "\n")  
            }  
        }  
    }  
}
```

STEP 8: Run Using Device Manager

1. Open Device Manager
2. Start Emulator
3. Click Run ►

OutPut



Ex. No. : 10

Date

Advanced RecyclerView

Aim

To implement an Advanced RecyclerView in Android using Kotlin, which displays a list of items and handles item click events.

Software Requirements

- Android Studio Otter
- Language: Kotlin
- Minimum SDK: API 21
- Target SDK: 33 or 34

Step 1: Create a New Project (VERY IMPORTANT)

1. Open Android Studio Otter
2. Click New Project
3. Select Empty Views Activity
4. Click Next
5. Fill details:
 - Name: RecyclerViewAdvanced
 - Language: Kotlin
 - Minimum SDK: API 21
6. Click Finish
7. Wait until Gradle Build finishes completely 

Step 2: Add RecyclerView Dependency

Open build.gradle (Module: app)

Add this inside dependencies { }

implementation "androidx.recyclerview:recyclerview:1.3.2"

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<androidx.constraintlayout.widget.ConstraintLayout
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent">
```

```
    <androidx.recyclerview.widget.RecyclerView
```

```
        android:id="@+id/recyclerView"
```

```
        android:layout_width="0dp"
```

```
        android:layout_height="0dp"
```

```
        app:layout_constraintTop_toTopOf="parent"
```

```
        app:layout_constraintBottom_toBottomOf="parent"
```

```
        app:layout_constraintStart_toStartOf="parent"
```

```
        app:layout_constraintEnd_toEndOf="parent"/>
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Step 4: Create Row Layout

res/layout/item_row.xml

```
<?xml version="1.0" encoding="utf-8"?>

<TextView

    xmlns:android="http://schemas.android.com/apk/res/android"

    android:id="@+id/txtItem"

    android:layout_width="match_parent"

    android:layout_height="wrap_content"

    android:padding="20dp"

    android:textSize="18sp"

    android:textColor="#000000"/>
```

Step 5: Create Data Model

Right click package → New → Kotlin Class/File

Name: Item.kt

```
data class Item(val name: String)
```

Step 6: Create RecyclerView Adapter

ItemAdapter.kt

```
import android.view.LayoutInflater

import android.view.View

import android.view.ViewGroup

import android.widget.TextView

import android.widget.Toast

import androidx.recyclerview.widget.RecyclerView

class ItemAdapter(private val itemList: List<Item>) :

    RecyclerView.Adapter<ItemAdapter.ItemViewHolder>() {
```

```

class ItemViewHolder(itemView: View) : RecyclerView.ViewHolder(itemView) {

    val textView: TextView = itemView.findViewById(R.id.txtItem)

}

override fun onCreateViewHolder(parent: ViewGroup, viewType: Int): ItemViewHolder {

    val view = LayoutInflater.from(parent.context)

        .inflate(R.layout.item_row, parent, false)

    return ItemViewHolder(view)

}

override fun onBindViewHolder(holder: ItemViewHolder, position: Int) {

    holder.textView.text = itemList[position].name

    holder.itemView.setOnClickListener {

        Toast.makeText(

            holder.itemView.context,

            "Clicked: ${itemList[position].name}",

            Toast.LENGTH_SHORT

        ).show()

    }

}

override fun getItemCount(): Int {

    return itemList.size

}

```

```

}

```

MainActivity.kt

```
import android.os.Bundle

import androidx.appcompat.app.AppCompatActivity

import androidx.recyclerview.widget.LinearLayoutManager

import androidx.recyclerview.widget.RecyclerView

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)

        val recyclerView = findViewById<RecyclerView>(R.id.recyclerView)

        recyclerView.layoutManager = LinearLayoutManager(this)

        val items = listOf(

            Item("Android"),

            Item("Kotlin"),

            Item("RecyclerView"),

            Item("Advanced Concepts"),

            Item("Click Handling")

        )

        val adapter = ItemAdapter(items)

        recyclerView.adapter = adapter

    }
}
```

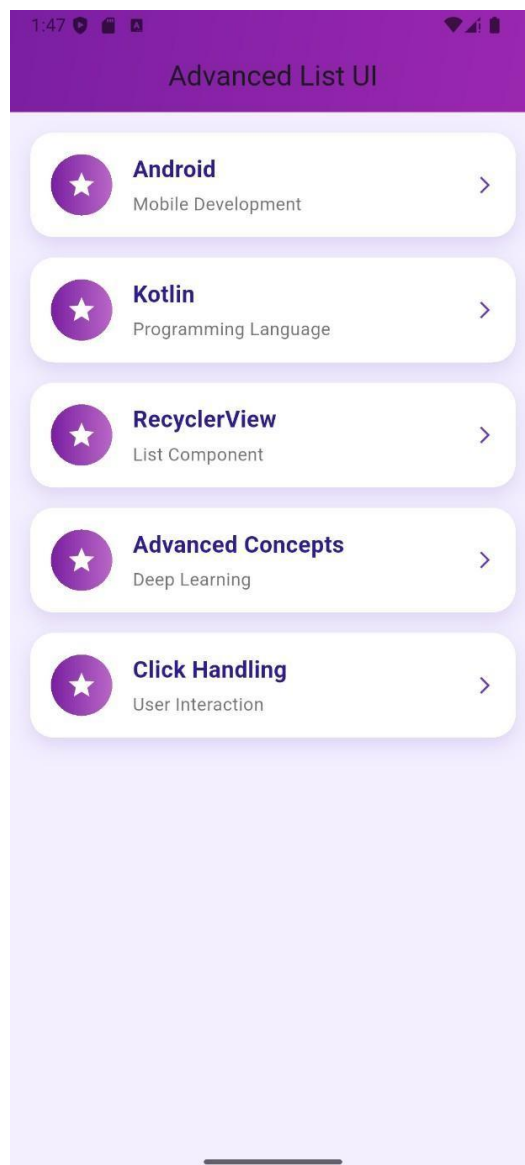


```
}
```

Step 8: Run the App

1. Select Emulator or Mobile Device
2. Click ► Run
3. Output:
 - List appears
 - Click any item → Toast message shown

Output :



Experiment No. 11 – Connect to the Internet (Android – Kotlin)

Aim

To develop an Android application that connects to the Internet and loads a web page using WebView.

Software / Tools Required

- Android Studio Otter
 - Language: Kotlin
 - Minimum SDK: API 21
 - Internet connection
-
- Android apps cannot access the Internet by default
 - Permission must be declared in AndroidManifest.xml
 - WebView is used to load and display web content inside an app

Procedure

Step 1: Create a New Project

1. Open Android Studio
2. New Project → Empty Views Activity
3. Name: InternetConnectApp
4. Language: Kotlin
5. Minimum SDK: API 21
6. Finish

Step 2: Add Internet Permission

AndroidManifest.xml

Add before <application> tag:

```
<uses-permission android:name="android.permission.INTERNET"/>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent"
```

```
    android:orientation="vertical">
```

```
    <WebView
```

```
        android:id="@+id/webView"
```

```
        android:layout_width="match_parent"
```

```
        android:layout_height="match_parent"/>
```

```
</LinearLayout>
```

MainActivity.kt

```
package com.example.internetconnectapp
```

```
import android.os.Bundle
```

```
import android.webkit.WebView
```

```
import android.webkit.WebViewClient
```

```
import androidx.appcompat.app.AppCompatActivity
```

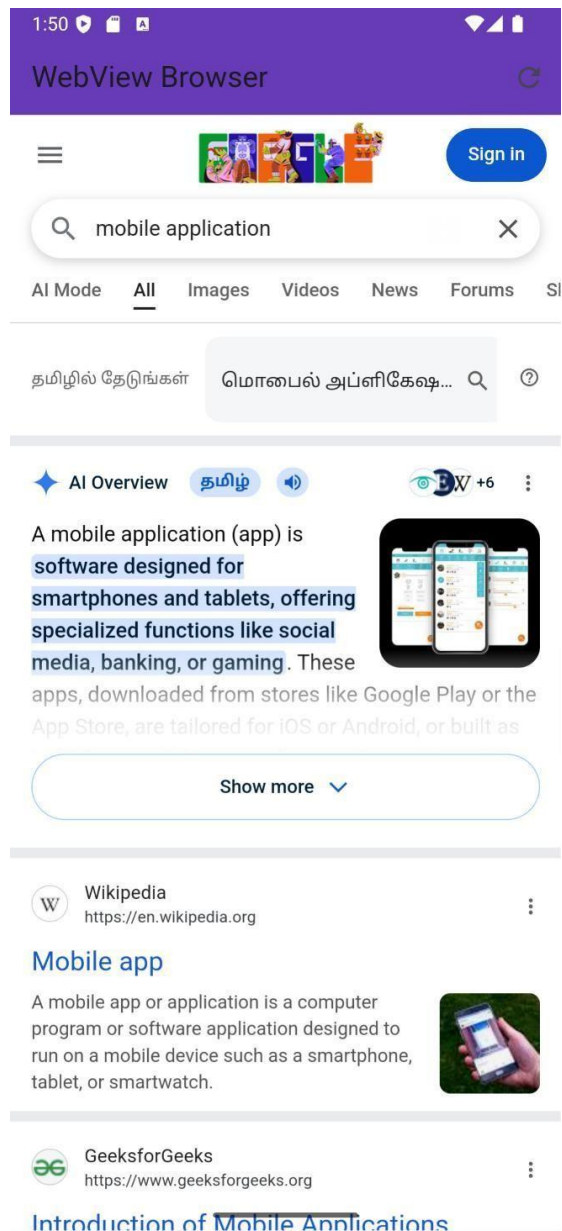
```
class MainActivity : AppCompatActivity() {
```

```
override fun onCreate(savedInstanceState: Bundle?) {  
  
    super.onCreate(savedInstanceState)  
  
    setContentView(R.layout.activity_main)  
  
  
    val webView = findViewById<WebView>(R.id.webView)  
  
  
    webView.webViewClient = WebViewClient()  
  
    webView.settings.javaScriptEnabled = true  
  
    webView.loadUrl("https://www.google.com")  
  
}  
}
```

Step 5: Run the Application

1. Connect emulator or mobile device
2. Click Run ►
3. Output:
 - Google page loads inside the app

Output :



Experiment No. 12

Implementation of Repository Pattern with Work Manager in Android

Aim

To implement the Repository Pattern for data handling and use WorkManager to perform a background task in an Android application.

Objectives

- To understand clean architecture using Repository Pattern
- To learn background task execution using WorkManager
- To integrate Repository with WorkManager

Repository Pattern

Repository Pattern is an architectural pattern that acts as a **single source of truth** for data. It separates **data access logic** from UI logic and is commonly used in **MVVM architecture**.

WorkManager

WorkManager is a Jetpack library used to perform **reliable background tasks**, even if the app is closed or the device restarts.

Step 1: Create a New Project

1. Open Android Studio
2. New Project → Empty Views Activity
3. Name: RepoWorkManagerApp
4. Language: Kotlin
5. Minimum SDK: API 21
6. Finish

Step 2: Add WorkManager Dependency

build.gradle (Module: app)

```
implementation "androidx.work:work-runtime-ktx:2.9.0"
```

Step 3: Create Worker Class

📁 Right click package → New → Kotlin Class

Name: DataSyncWorker.kt

```
import android.content.Context
```

```
import androidx.work.Worker
```

```
import androidx.work.WorkerParameters
```

```
class DataSyncWorker(  
    context: Context,  
    params: WorkerParameters  
) : Worker(context, params) {  
  
    override fun doWork(): Result {  
        println("Data syncing in background")  
        return Result.success()  
    }  
}
```

Step 4: Create Repository Class

📁 New → Kotlin Class

Name: DataRepository.kt

```
import android.content.Context
```

```
import androidx.work.OneTimeWorkRequestBuilder
```

```
import androidx.work.WorkManager
```

```
class DataRepository(private val context: Context) {
```

```
fun syncData() {  
    val workRequest =  
        OneTimeWorkRequestBuilder<DataSyncWorker>().build()  
  
    WorkManager.getInstance(context).enqueue(workRequest)  
}  
}
```

Step 5: Update UI Layout

➤ **res/layout/activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>  
  
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:gravity="center"  
    android:orientation="vertical">  
  
    <Button  
        android:id="@+id/btnSync"  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:text="Sync Data"/>  
  
</LinearLayout>
```


Step 6: MainActivity Code

MainActivity.kt

```
import android.os.Bundle

import android.widget.Button

import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {

        super.onCreate(savedInstanceState)

        setContentView(R.layout.activity_main)

        val repository = DataRepository(this)

        val button = findViewById<Button>(R.id.btnSync)

        button.setOnClickListener {

            repository.syncData()

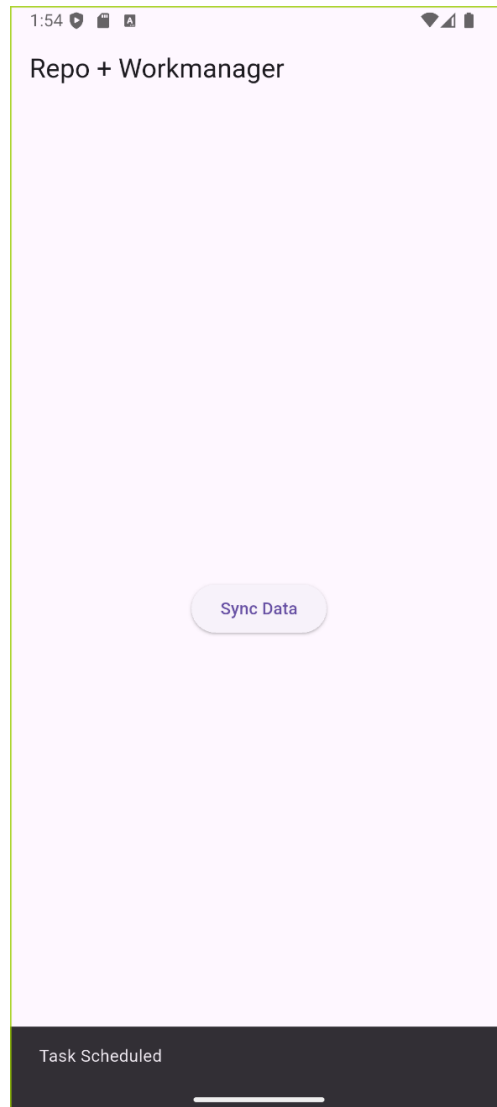
        }

    }

}
```

Output

- **When the Sync Data button is clicked**
- **Background task executes using WorkManager**
- **Data syncing message appears in Logcat**



Experiment No. 13 – App UI Design (Android – Kotlin)

Designing a User Interface (UI) in Android using Layouts, Widgets, and Views

Aim

To design an interactive and user-friendly Android application UI using various layouts, widgets, and views.

Objectives

- Understand different layouts in Android (Linear, Relative, Constraint, Frame)
- Learn to use widgets like Button, TextView, EditText, ImageView
- Implement interactive UI components
- Make the UI visually appealing and responsive

- **Layout:** Defines how UI components are arranged
- **LinearLayout:** Arranges views vertically or horizontally
- **RelativeLayout:** Positions views relative to others
- **ConstraintLayout:** Flexible layout for modern UI design
- **Widgets / Views:** Interactive components like Button, TextView, EditText, ImageView
- **Styles / Themes:** Control color, font, and overall look of the app

Step 1: Create a New Project

1. Open Android Studio → New Project → Empty Activity
2. Name: AppUIDesign
3. Language: Kotlin
4. Minimum SDK: API 21
5. Finish

Step 2: Open Layout File

- Open res/layout/activity_main.xml
- Switch to Design View (drag-and-drop) or Code View (XML)

Step 3: Add Layout

Example **ConstraintLayout**:

```
<androidx.constraintlayout.widget.ConstraintLayout
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:app="http://schemas.android.com/apk/res-auto"
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent">
```

```
<TextView
```

```
    android:id="@+id/txtTitle"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="Welcome to My App"
```

```
    android:textSize="24sp"
```

```
    app:layout_constraintTop_toTopOf="parent"
```

```
    app:layout_constraintStart_toStartOf="parent"
```

```
    app:layout_constraintEnd_toEndOf="parent"/>
```

```
<EditText
```

```
    android:id="@+id/edtName"
```

```
    android:layout_width="0dp"
```

```
    android:layout_height="wrap_content"
```

```
    android:hint="Enter your name"
```

```
app:layout_constraintTop_toBottomOf="@id/txtTitle"
```

```
app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintEnd_toEndOf="parent"
```

```
android:layout_margin="16dp"/>
```

```
<Button
```

```
    android:id="@+id/btnSubmit"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="Submit"
```

```
    app:layout_constraintTop_toBottomOf="@id/edtName"
```

```
    app:layout_constraintStart_toStartOf="parent"
```

```
    app:layout_constraintEnd_toEndOf="parent"
```

```
    android:layout_marginTop="16dp"/>
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

```
MainActivity.kt
```

```
package com.example.appuidesign
```

```
import android.os.Bundle
```

```
import android.widget.Button
```

```
import android.widget.EditText
```

```
import android.widget.TextView
```

```
import android.widget.Toast
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        setContentView(R.layout.activity_main)  
        val edtName = findViewById<EditText>(R.id.edtName)  
        val btnSubmit = findViewById<Button>(R.id.btnSubmit)  
        val txtTitle = findViewById<TextView>(R.id.txtTitle)  
        btnSubmit.setOnClickListener {  
            val name = edtName.text.toString()  
            if (name.isNotEmpty()) {  
                Toast.makeText(this, "Hello, $name!", Toast.LENGTH_SHORT).show()  
                txtTitle.text = "Welcome, $name"  
            } else {  
                Toast.makeText(this, "Please enter your name", Toast.LENGTH_SHORT).show()  
            }  
        }  
    }  
}
```

Step 5: Run the Application

1. Connect Emulator / Device
2. Click Run ►
3. Output:
 - TextView displays welcome text
 - EditText accepts input
 - Button click shows Toast and updates TextView

Output :

